

SAFETY DATA SHEET

Emit 2000 Valproic Acid Assay

SDS no.:

SY4G019

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name : Emit 2000 Valproic Acid Assay
Product code : 4G019UL, OSR4G229, RG329, RG359, 10445312, 10445314, 10485118, 10485119, 06592706

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses Emit® 2000 Valproic Acid Assay, Diagnostic agents.
Reagent 1
Emit® 2000 Valproic Acid Assay, Diagnostic agents.
Reagent 2
Restrictions on use For professional users only.

1.3 Details of the supplier of the safety data sheet

Manufactured/supplied : Siemens Healthcare Diagnostics Limited
Park View,
Watchmoor Park,
Camberley,
Surrey,
GU15 3YL
United Kingdom

Phone: +44 (0) 345 600 1955
e-mail address of person responsible for this SDS : dx.msds.healthcare@siemens-healthineers.com

1.4 Emergency telephone number

Poison Control:
In England and Wales:
NHS Direct – 0845 4647 or 111
In Scotland: NHS 24 – 08454 24 24 24
In the Republic of Ireland: 01 809 2166

CHEMTREC: 0870-8200418 (UK only)
+(353)-19014670 (UK & Ireland)
(International calls to the United Kingdom)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Product definition : Emit® 2000 Valproic Acid Assay, Mixture
Reagent 1
Emit® 2000 Valproic Acid Assay, Mixture
Reagent 2

Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Emit® 2000 Valproic Acid Assay, Reagent 1

Skin Sens. 1, H317
Aquatic Chronic 3, H412

Emit® 2000 Valproic Acid Assay, Reagent 1

The product is classified as hazardous according to Regulation (EC) 1272/2008 as amended.

Emit® 2000 Valproic Acid Assay, Reagent 2

The product is not classified as hazardous according to Regulation (EC) 1272/2008 as amended.

See Section 16 for the full text of the H statements declared above.

See Section 11 for more detailed information on health effects and symptoms.

Emit 2000 Valproic Acid Assay

SECTION 2: Hazards identification**2.2 Label elements****Hazard pictograms**

:

**Signal word**: Emit® 2000 Valproic Acid Assay, Reagent 1
Emit® 2000 Valproic Acid Assay, Reagent 2

Warning

No signal word.

Hazard statements

: Emit® 2000 Valproic Acid Assay, Reagent 1

H317 - May cause an allergic skin reaction.
H412 - Harmful to aquatic life with long lasting effects.
No known significant effects or critical hazards.

Emit® 2000 Valproic Acid Assay, Reagent 2

Precautionary statements**Prevention**

: Emit® 2000 Valproic Acid Assay, Reagent 1

P280 - Wear protective gloves/ protective clothing/eye protection/face protection.
P273 - Avoid release to the environment.
Not applicable.

Emit® 2000 Valproic Acid Assay, Reagent 2

Response

: Emit® 2000 Valproic Acid Assay, Reagent 1

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water.
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention.
P362 + P364 - Take off contaminated clothing and wash it before reuse.
Not applicable.

Emit® 2000 Valproic Acid Assay, Reagent 2

Storage: Emit® 2000 Valproic Acid Assay, Reagent 1
Emit® 2000 Valproic Acid Assay, Reagent 2

Not applicable.

Not applicable.

Disposal

: Emit® 2000 Valproic Acid Assay, Reagent 1

P501 - Dispose of contents and container in accordance with all local, regional, and national regulations.
Not applicable.

Emit® 2000 Valproic Acid Assay, Reagent 2

Hazardous ingredients: **Emit® 2000 Valproic Acid Assay, Reagent 1**
reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1)**Supplemental label elements**: Emit® 2000 Valproic Acid Assay, Reagent 1
Emit® 2000 Valproic Acid Assay, Reagent 2

Not applicable.

Not applicable.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles: Emit® 2000 Valproic Acid Assay, Reagent 1
Emit® 2000 Valproic Acid Assay, Reagent 2

Not applicable.

Not applicable.

2.3 Other hazards**Product meets the criteria for PBT or vPvB according to Regulation (EC) No. 1907/2006, Annex XIII**

: This mixture does not contain any substances that are assessed to be a PBT or a vPvB.

Emit 2000 Valproic Acid Assay

SECTION 2: Hazards identification

Other hazards which do not result in classification : Emit® 2000 Valproic Acid Assay, Reagent 1 None known.
 Emit® 2000 Valproic Acid Assay, Reagent 2 None known.

Additional information : Not available.

Sodium azide may react with lead or copper plumbing to form highly explosive metal azides.

SECTION 3: Composition/information on ingredients

3.2 Mixtures : Emit® 2000 Valproic Acid Assay, Mixture
 Reagent 1
 Emit® 2000 Valproic Acid Assay, Mixture
 Reagent 2

Product/ingredient name	Identifiers	%	Classification	Specific Conc. Limits, M-factors and ATEs	Type
Emit® 2000 Valproic Acid Assay, Reagent 1 reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1)	CAS: 55965-84-9 Index: 613-167-00-5	<0.025	Acute Tox. 3, H301 Acute Tox. 2, H310 Acute Tox. 2, H330 Skin Corr. 1C, H314 Skin Sens. 1A, H317 Aquatic Acute 1, H400 Aquatic Chronic 1, H410 EUH071 See Section 16 for the full text of the H statements declared above.	ATE [Oral] = 53 mg/kg ATE [Dermal] = 50 mg/kg ATE [Inhalation (vapours)] = 0.5 mg/l Skin Corr. 1C, H314: C ≥ 0.6% Skin Irrit. 2, H315: 0.06% ≤ C < 0.6% Skin Sens. 1, H317: C ≥ 0.0015% M [Acute] = 100 M [Chronic] = 100	[1]

Type

[1] Substance classified with a health or environmental hazard

Occupational exposure limits, if available, are listed in Section 8.

SECTION 4: First aid measures**4.1 Description of first aid measures**

Eye contact : Emit® 2000 Valproic Acid Assay, Reagent 1 Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Get medical attention if irritation occurs.
 Emit® 2000 Valproic Acid Assay, Reagent 2 Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Get medical attention if irritation occurs.

Emit 2000 Valproic Acid Assay

SECTION 4: First aid measures

Inhalation	: Emit® 2000 Valproic Acid Assay, Reagent 1	Remove victim to fresh air and keep at rest in a position comfortable for breathing. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention if adverse health effects persist or are severe. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical attention if symptoms occur. In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
Skin contact	: Emit® 2000 Valproic Acid Assay, Reagent 1	Wash with plenty of soap and water. Remove contaminated clothing and shoes. Wash contaminated clothing thoroughly with water before removing it, or wear gloves. Continue to rinse for at least 10 minutes. Get medical attention. In the event of any complaints or symptoms, avoid further exposure. Wash clothing before reuse. Clean shoes thoroughly before reuse.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Flush contaminated skin with plenty of water. Remove contaminated clothing and shoes. Get medical attention if symptoms occur.
Ingestion	: Emit® 2000 Valproic Acid Assay, Reagent 1	Wash out mouth with water. Remove dentures if any. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Get medical attention if adverse health effects persist or are severe. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Wash out mouth with water. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting unless directed to do so by medical personnel. Get medical attention if symptoms occur.

Emit 2000 Valproic Acid Assay

SECTION 4: First aid measures

Protection of first-aiders	: Emit® 2000 Valproic Acid Assay, Reagent 1	No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No action shall be taken involving any personal risk or without suitable training.

4.2 Most important symptoms and effects, both acute and delayed**Over-exposure signs/symptoms**

Eye contact	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific data.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No specific data.
Inhalation	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific data.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No specific data.
Skin contact	: Emit® 2000 Valproic Acid Assay, Reagent 1	Adverse symptoms may include the following: irritation redness
	Emit® 2000 Valproic Acid Assay, Reagent 2	No specific data.
Ingestion	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific data.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician	: Emit® 2000 Valproic Acid Assay, Reagent 1	In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
	Emit® 2000 Valproic Acid Assay, Reagent 2	In case of inhalation of decomposition products in a fire, symptoms may be delayed. The exposed person may need to be kept under medical surveillance for 48 hours.
Specific treatments	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific treatment.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No specific treatment.

SECTION 5: Firefighting measures**5.1 Extinguishing media**

Suitable extinguishing media	: Use an extinguishing agent suitable for the surrounding fire.
Unsuitable extinguishing media	: None known.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture	: In a fire or if heated, a pressure increase will occur and the container may burst. This material is harmful to aquatic life with long lasting effects. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
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Emit 2000 Valproic Acid Assay

SECTION 5: Firefighting measures

Hazardous combustion products : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
nitrogen oxides
phosphorus oxides

5.3 Advice for firefighters

Special protective actions for fire-fighters : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilt material. Avoid breathing vapour or mist. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment.

For emergency responders : If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

6.2 Environmental precautions

: Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.

6.3 Methods and material for containment and cleaning up

Small spill : Stop leak if without risk. Move containers from spill area. Dilute with water and mop up if water-soluble. Alternatively, or if water-insoluble, absorb with an inert dry material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.

Large spill : Stop leak if without risk. Move containers from spill area. Approach the release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilt product.

6.4 Reference to other sections

: See Section 1 for emergency contact information.
See Section 8 for information on appropriate personal protective equipment.
See Section 13 for additional waste treatment information.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Protective measures : Put on appropriate personal protective equipment (see Section 8). Persons with a history of skin sensitization problems should not be employed in any process in which this product is used. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing vapour or mist. Avoid release to the environment. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be hazardous. Do not reuse container.

Emit 2000 Valproic Acid Assay

SECTION 7: Handling and storage

Advice on general occupational hygiene : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2 Conditions for safe storage, including any incompatibilities

Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

7.3 Specific end use(s)

Recommendations : Not available.

Industrial sector specific solutions : Not available.

SECTION 8: Exposure controls/personal protection

The information in this section contains generic advice and guidance. Information is provided based on typical anticipated uses of the product. Additional measures might be required for bulk handling or other uses that could significantly increase worker exposure or environmental releases.

8.1 Control parameters**Occupational exposure limits**

No exposure limit value known.

Recommended monitoring procedures : If this product contains ingredients with exposure limits, personal, workplace atmosphere or biological monitoring may be required to determine the effectiveness of the ventilation or other control measures and/or the necessity to use respiratory protective equipment. Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be required.

DNELs/DMELs

Product/ingredient name	Type	Exposure	Value	Population	Effects
Emit® 2000 Valproic Acid Assay, Reagent 1 reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1)	DNEL	Long term Inhalation	0.02 mg/m ³	General population	Local
	DNEL	Long term Inhalation	0.02 mg/m ³	Workers	Local
	DNEL	Short term Inhalation	0.04 mg/m ³	General population	Local
	DNEL	Short term Inhalation	0.04 mg/m ³	Workers	Local
	DNEL	Long term Oral	0.09 mg/kg bw/day	General population	Systemic
	DNEL	Short term Oral	0.11 mg/kg bw/day	General population	Systemic

PNECs

No PNECs available

Emit 2000 Valproic Acid Assay

SECTION 8: Exposure controls/personal protection**8.2 Exposure controls**

Appropriate engineering controls : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Individual protection measures

Hygiene measures : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Eye/face protection : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: safety glasses with side-shields.

Skin protection

Hand protection : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated.

Impervious gloves (e.g. butyl, nitrile, etc.) are recommended if skin contact is possible and for processing operations. Protective gloves must meet the standards in accordance with CEN EN374, ASTM F1001 or international equivalent.

Body protection : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Other skin protection : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Respiratory protection : Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.

Environmental exposure controls : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

9.1 Information on basic physical and chemical properties**Appearance**

Physical state	: Emit® 2000 Valproic Acid Assay, Reagent 1	Liquid.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Liquid.
Colour	: Emit® 2000 Valproic Acid Assay, Reagent 1	Clear. Yellowish.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Clear. Yellowish.
Odour	: Emit® 2000 Valproic Acid Assay, Reagent 1	Odourless.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Odourless.

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SECTION 9: Physical and chemical properties

Odour threshold	: Not relevant/applicable due to nature of the product.	
Melting point/freezing point	: Not relevant/applicable due to nature of the product.	
Softening point	: Not relevant/applicable due to nature of the product.	
Sublimation temperature	: Not relevant/applicable due to nature of the product.	
Initial boiling point and boiling range	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Flammability	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not relevant/applicable due to nature of the product.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not relevant/applicable due to nature of the product.
Lower and upper explosion limit	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Flash point	: Emit® 2000 Valproic Acid Assay, Reagent 1	[Product does not sustain combustion.]
	Emit® 2000 Valproic Acid Assay, Reagent 2	[Product does not sustain combustion.]

Ingredient name	Closed cup			Open cup		
	°C	°F	Method	°C	°F	Method
Emit® 2000 Valproic Acid Assay, Reagent 1 triethyl orthoformate	35	95				
Emit® 2000 Valproic Acid Assay, Reagent 2 methanol	9.7	49.5	Abel-Pensky			

Auto-ignition temperature : Not relevant/applicable due to nature of the product.

Ingredient name	°C	°F	Method
Emit® 2000 Valproic Acid Assay, Reagent 1 1,1'-oxydipropyl-2-ol	310	590	
Emit® 2000 Valproic Acid Assay, Reagent 2 sodium azide	309	588.2	EU A.16

Decomposition temperature : Not relevant/applicable due to nature of the product.

pH	: Emit® 2000 Valproic Acid Assay, Reagent 1	6 [Conc. (% w/w): 1%]
	Emit® 2000 Valproic Acid Assay, Reagent 2	7.6 [Conc. (% w/w): 1%]
Viscosity	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Solubility(ies) :

Media	Result
Emit® 2000 Valproic Acid Assay, Reagent 2 cold water	Easily soluble
hot water	Easily soluble

Solubility in water : Not relevant/applicable due to nature of the product.**Miscible with water** : Not relevant/applicable due to nature of the product.

Emit 2000 Valproic Acid Assay

SECTION 9: Physical and chemical properties

Partition coefficient: n-octanol/ water : Not relevant/applicable due to nature of the product.

Vapour pressure :

Ingredient name	Vapour Pressure at 20°C			Vapour pressure at 50°C		
	mm Hg	kPa	Method	mm Hg	kPa	Method
Emit® 2000 Valproic Acid Assay, Reagent 1 water	23.8	3.2				
Emit® 2000 Valproic Acid Assay, Reagent 2 methanol	126.96	16.9				

Evaporation rate : Not relevant/applicable due to nature of the product.

Relative density : Emit® 2000 Valproic Acid Assay, Reagent 1 1.012
Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Density : Emit® 2000 Valproic Acid Assay, Reagent 1 1.008 g/cm³
Emit® 2000 Valproic Acid Assay, Reagent 2 1.041 g/cm³

Vapour density : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Explosive properties : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Oxidising properties : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Particle characteristics

Median particle size : Not applicable.

9.2 Other information

Fire point : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Burning time : Not relevant/applicable due to nature of the product.

Fundamental burning velocity : Not relevant/applicable due to nature of the product.

Burning rate : Not relevant/applicable due to nature of the product.

SADT : Not relevant/applicable due to nature of the product.

SAPT : Not relevant/applicable due to nature of the product.

Heat of reaction : Not relevant/applicable due to nature of the product.

Heat of combustion : Not relevant/applicable due to nature of the product.

Flow time (ISO 2431) : Not relevant/applicable due to nature of the product.

Molecular weight : Emit® 2000 Valproic Acid Assay, Reagent 1 Not applicable.
Emit® 2000 Valproic Acid Assay, Reagent 2 Not applicable.

Emit 2000 Valproic Acid Assay

SECTION 10: Stability and reactivity

10.1 Reactivity	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific test data related to reactivity available for this product or its ingredients.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	No specific test data related to reactivity available for this product or its ingredients.
10.2 Chemical stability	: Emit® 2000 Valproic Acid Assay, Reagent 1	The product is stable.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	The product is stable.
10.3 Possibility of hazardous reactions	: Emit® 2000 Valproic Acid Assay, Reagent 1	Under normal conditions of storage and use, hazardous reactions will not occur.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Under normal conditions of storage and use, hazardous reactions will not occur.
10.4 Conditions to avoid	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific data.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	No specific data.
10.5 Incompatible materials	: Emit® 2000 Valproic Acid Assay, Reagent 1	No specific data.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	No specific data.
10.6 Hazardous decomposition products	: Emit® 2000 Valproic Acid Assay, Reagent 1	Under normal conditions of storage and use, hazardous decomposition products should not be produced.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information**11.1 Information on toxicological effects****Acute toxicity**

Product/ingredient name	Result	Species	Dose	Exposure
Emit® 2000 Valproic Acid Assay, Reagent 1 reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1)	LD50 Oral	Rat	53 mg/kg	-

Conclusion/Summary	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Acute toxicity estimates

Product/ingredient name	Oral (mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapours) (mg/l)	Inhalation (dusts and mists) (mg/l)
Emit® 2000 Valproic Acid Assay, Reagent 1 reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1)	53	50	N/A	0.5	N/A

Emit 2000 Valproic Acid Assay

SECTION 11: Toxicological information**Irritation/Corrosion**

Product/ingredient name	Result	Species	Score	Exposure	Observation
Emit® 2000 Valproic Acid Assay, Reagent 1 reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1)	Skin - Severe irritant	Human	-	0.01 %	-

Conclusion/Summary

Skin	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Eyes	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Respiratory	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Sensitisation**Conclusion/Summary**

Skin	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Respiratory	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Mutagenicity**Conclusion/Summary**

: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Carcinogenicity**Conclusion/Summary**

: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Reproductive toxicity**Conclusion/Summary**

: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Teratogenicity**Conclusion/Summary**

: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

Specific target organ toxicity (single exposure)

Not available.

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Emit 2000 Valproic Acid Assay

SECTION 11: Toxicological information

Not available.

Information on likely routes of exposure : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
 Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Potential acute health effects

Eye contact : Emit® 2000 Valproic Acid Assay, Reagent 1 No known significant effects or critical hazards.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No known significant effects or critical hazards.

Inhalation : Emit® 2000 Valproic Acid Assay, Reagent 1 No known significant effects or critical hazards.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No known significant effects or critical hazards.

Skin contact : Emit® 2000 Valproic Acid Assay, Reagent 1 May cause an allergic skin reaction.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No known significant effects or critical hazards.

Ingestion : Emit® 2000 Valproic Acid Assay, Reagent 1 No known significant effects or critical hazards.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No known significant effects or critical hazards.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact : Emit® 2000 Valproic Acid Assay, Reagent 1 No specific data.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No specific data.

Inhalation : Emit® 2000 Valproic Acid Assay, Reagent 1 No specific data.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No specific data.

Skin contact : Emit® 2000 Valproic Acid Assay, Reagent 1 Adverse symptoms may include the following:
 irritation
 redness
 Emit® 2000 Valproic Acid Assay, Reagent 2 No specific data.

Ingestion : Emit® 2000 Valproic Acid Assay, Reagent 1 No specific data.
 Emit® 2000 Valproic Acid Assay, Reagent 2 No specific data.

Delayed and immediate effects as well as chronic effects from short and long-term exposure**Short term exposure**

Potential immediate effects : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
 Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Potential delayed effects : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
 Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

Long term exposure

Potential immediate effects : Emit® 2000 Valproic Acid Assay, Reagent 1 Not available.
 Emit® 2000 Valproic Acid Assay, Reagent 2 Not available.

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SECTION 11: Toxicological information

Potential delayed effects	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
<u>Potential chronic health effects</u>		
Not available.		
Conclusion/Summary	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
General	: Emit® 2000 Valproic Acid Assay, Reagent 1	Once sensitized, a severe allergic reaction may occur when subsequently exposed to very low levels.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	No known significant effects or critical hazards.
Carcinogenicity	: Emit® 2000 Valproic Acid Assay, Reagent 1	No known significant effects or critical hazards.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	No known significant effects or critical hazards.
Mutagenicity	: Emit® 2000 Valproic Acid Assay, Reagent 1	No known significant effects or critical hazards.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	No known significant effects or critical hazards.
Reproductive toxicity	: No known significant effects or critical hazards.	
Interactive effects	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
<u>Toxicokinetics</u>		
Absorption	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Distribution	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Metabolism	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Elimination	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

11.2 Information on other hazards**11.2.1 Endocrine disrupting properties**

Not available.

11.2.2 Other information

Emit® 2000 Valproic Acid Assay, Reagent 1 : Not available.

Emit® 2000 Valproic Acid Assay, Reagent 2 : Not available.

Emit 2000 Valproic Acid Assay

SECTION 12: Ecological information**12.1 Toxicity**

Conclusion/Summary	:	Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
		Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

12.2 Persistence and degradability

Conclusion/Summary	:	Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
		Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

12.3 Bioaccumulative potential

Not available.

12.4 Mobility in soil

Soil/water partition coefficient (K_{oc})	:	Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
		Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.
Mobility	:	Emit® 2000 Valproic Acid Assay, Reagent 1	Not available.
		Emit® 2000 Valproic Acid Assay, Reagent 2	Not available.

12.5 Results of PBT and vPvB assessment

This mixture does not contain any substances that are assessed to be a PBT or a vPvB.

12.6 Endocrine disrupting properties

Not available.

12.7 Other adverse effects

Emit® 2000 Valproic Acid Assay, Reagent 1	No known significant effects or critical hazards.
Emit® 2000 Valproic Acid Assay, Reagent 2	No known significant effects or critical hazards.

SECTION 13: Disposal considerations**13.1 Waste treatment methods****Product**

Methods of disposal	:	The generation of waste should be avoided or minimised wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction.
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Hazardous waste	:	<table border="0"> <tr> <td>Emit® 2000 Valproic Acid Assay, Reagent 1</td> <td>The classification of the product may meet the criteria for a hazardous waste.</td> </tr> <tr> <td>Emit® 2000 Valproic Acid Assay, Reagent 2</td> <td>Within the present knowledge of the supplier, this product is not regarded as hazardous waste, as defined by EU Directive 2008/98/EC.</td> </tr> </table>	Emit® 2000 Valproic Acid Assay, Reagent 1	The classification of the product may meet the criteria for a hazardous waste.	Emit® 2000 Valproic Acid Assay, Reagent 2	Within the present knowledge of the supplier, this product is not regarded as hazardous waste, as defined by EU Directive 2008/98/EC.
Emit® 2000 Valproic Acid Assay, Reagent 1	The classification of the product may meet the criteria for a hazardous waste.					
Emit® 2000 Valproic Acid Assay, Reagent 2	Within the present knowledge of the supplier, this product is not regarded as hazardous waste, as defined by EU Directive 2008/98/EC.					
		Sodium azide may react with lead or copper plumbing to form highly explosive metal azides.				

Packaging

Methods of disposal	:	The generation of waste should be avoided or minimised wherever possible. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible.
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Emit 2000 Valproic Acid Assay

SECTION 13: Disposal considerations

Special precautions : This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers.

SECTION 14: Transport information**ADR/RID**

14.1 UN number	Emit® 2000 Valproic Acid Assay, Reagent 1	Not regulated.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not regulated.
14.2 UN proper shipping name	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.3 Transport hazard class(es)	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.4 Packing group	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.5 Environmental hazards	Emit® 2000 Valproic Acid Assay, Reagent 1	No.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No.
Additional information	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-

ADN

14.1 UN number	Emit® 2000 Valproic Acid Assay, Reagent 1	Not regulated.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not regulated.
14.2 UN proper shipping name	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.3 Transport hazard class(es)	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.4 Packing group	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.5 Environmental hazards	Emit® 2000 Valproic Acid Assay, Reagent 1	No.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No.
Additional information	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-

IMDG

14.1 UN number	Emit® 2000 Valproic Acid Assay, Reagent 1	Not regulated.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not regulated.
14.2 UN proper shipping name	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.3 Transport hazard class(es)	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.4 Packing group	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-

Emit 2000 Valproic Acid Assay

SECTION 14: Transport information

14.5	Emit® 2000 Valproic Acid Assay, Reagent 1	No.
Environmental hazards	Emit® 2000 Valproic Acid Assay, Reagent 2	No.
Additional information	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-

IATA

14.1 UN number	Emit® 2000 Valproic Acid Assay, Reagent 1	Not regulated.
	Emit® 2000 Valproic Acid Assay, Reagent 2	Not regulated.
14.2 UN proper shipping name	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.3 Transport hazard class(es)	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-

14.4 Packing group	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-
14.5 Environmental hazards	Emit® 2000 Valproic Acid Assay, Reagent 1	No.
	Emit® 2000 Valproic Acid Assay, Reagent 2	No.
Additional information	Emit® 2000 Valproic Acid Assay, Reagent 1	-
	Emit® 2000 Valproic Acid Assay, Reagent 2	-

14.6 Special precautions for user : Emit® 2000 Valproic Acid Assay, Reagent 1

Emit® 2000 Valproic Acid Assay, Reagent 2

Transport within user's premises:
always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport within user's premises:
always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

14.7 Transport in bulk according to IMO instruments Not applicable.

Notes : A "-" = not applicable.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture
Regulation

EU Regulation (EC) No. 1907/2006 (REACH)**Annex XIV - List of substances subject to authorisation****Annex XIV**

None of the components are listed.

Substances of very high concern

None of the components are listed.

Emit 2000 Valproic Acid Assay

SECTION 15: Regulatory information

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

: Emit® 2000 Valproic Acid Assay, Reagent 1	Not applicable.
: Emit® 2000 Valproic Acid Assay, Reagent 2	Not applicable.

Other EU regulations

Industrial emissions (integrated pollution prevention and control) - Air	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not listed
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not listed
Industrial emissions (integrated pollution prevention and control) - Water	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not listed
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not listed

Ozone depleting substances (1005/2009/EU)

Not listed.

Prior Informed Consent (PIC) (649/2012/EU)

Not listed.

Persistent Organic Pollutants

Not listed.

Aerosol dispensers	: Emit® 2000 Valproic Acid Assay, Reagent 1	Not applicable.
	: Emit® 2000 Valproic Acid Assay, Reagent 2	Not applicable.

Seveso Directive

Emit® 2000 Valproic Acid Assay, Reagent 1

This product is not controlled under the Seveso Directive.

Emit® 2000 Valproic Acid Assay, Reagent 2

This product is not controlled under the Seveso Directive.

National regulations**International regulations****Montreal Protocol**

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

15.2 Chemical safety assessment : Not applicable.

SECTION 16: Other information

Indicates information that has changed from previously issued version.

Abbreviations and acronyms

: ATE = Acute Toxicity Estimate
: CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008]
: DMEL = Derived Minimal Effect Level
: DNEL = Derived No Effect Level
: EUH statement = CLP-specific Hazard statement
: PBT = Persistent, Bioaccumulative and Toxic
: PNEC = Predicted No Effect Concentration

Emit 2000 Valproic Acid Assay

SECTION 16: Other information

RRN = REACH Registration Number
 vPvB = Very Persistent and Very Bioaccumulative
 ASTM = American Society of Testing Materials
 CEN = European Committee on Standardization
 ECHA = European Chemicals Agency
 RTECS = Registry of Toxic Effects of Chemical Substances

Key literature references and sources for data : This SDS was prepared on the basis of sheets of individual components, literature data, online databases (e.g. ECHA, RTECS) as well as our knowledge and experience, taking into account current legislation.

Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Classification	Justification
Emit® 2000 Valproic Acid Assay, Reagent 1 Skin Sens. 1, H317 Aquatic Chronic 3, H412	Calculation method Calculation method

Full text of abbreviated H statements

**Emit® 2000
Valproic Acid
Assay,
Reagent 1**

H301 Toxic if swallowed.
 H310 Fatal in contact with skin.
 H314 Causes severe skin burns and eye damage.
 H317 May cause an allergic skin reaction.
 H330 Fatal if inhaled.
 H400 Very toxic to aquatic life.
 H410 Very toxic to aquatic life with long lasting effects.
 H412 Harmful to aquatic life with long lasting effects.
 EUH071 Corrosive to the respiratory tract.

Full text of classifications [CLP/GHS]

**Emit® 2000
Valproic Acid Assay,
Reagent 1**

Acute Tox. 2 ACUTE TOXICITY - Category 2
 Acute Tox. 3 ACUTE TOXICITY - Category 3
 Aquatic Acute 1 SHORT-TERM (ACUTE) AQUATIC HAZARD - Category 1
 Aquatic Chronic 1 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 1
 Aquatic Chronic 3 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 3
 Skin Corr. 1C SKIN CORROSION/IRRITATION - Category 1C
 Skin Sens. 1 SKIN SENSITISATION - Category 1
 Skin Sens. 1A SKIN SENSITISATION - Category 1A

Training advice : Provide workers with adequate training to assure that chemicals are handled safely in accordance with national and community legislation.

Date of printing : 12/11/2022

Date of issue/ Date of revision : 12/11/2022

Date of previous issue : No previous validation

Version : 1

Notice to reader

To the best of our knowledge, the information contained herein is accurate. However, neither the above-named supplier, nor any of its subsidiaries, assumes any liability whatsoever for the accuracy or completeness of the information contained herein.

Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards that exist.

Emit 2000 Valproic Acid Assay